

Supervised Learning Foundations

Executive Brief

Decision guidance for responsible predictive model development

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Supervised learning becomes useful when an institution can show what was predicted, what information was available, how performance was evaluated, and who remains accountable for action. For the controlled service-demand case, the learned model improves test RMSE from 40.496 for the seasonal baseline to 30.876, while a leaked result of 2.048 is explicitly rejected.

Decision message

The controlled evidence supports incremental predictive value in the synthetic laboratory. It does not yet authorize operational deployment. Progression to use requires confirmation of data provenance, temporal validity, uncertainty, subgroup performance, capacity effects, monitoring and accountable review.

Controlled results

Evidence	Result	Interpretation
Training mean baseline	RMSE 103.958	Weak constant reference
Seasonal naive baseline	RMSE 40.496	Credible operational comparator
Learned regression model	RMSE 30.876	23.8% improvement versus seasonal
Leaked model	RMSE 2.048	Rejected; post-outcome information
High-demand classifier	ROC AUC 0.955	Strong ranking; threshold still governed

Controls before deployment

- Freeze the prediction time and admissible feature list
- Use temporal or otherwise dependence-aware evaluation
- Compare the same test cases against credible baselines
- Quantify uncertainty and examine consequential subgroups
- Define warning, suspension and recovery thresholds
- Keep observed facts, model estimates and authorized decisions distinct

Governance questions

Question	Required record
What decision is supported?	Purpose, user, authority and prohibited uses
What can the model know?	Feature provenance and availability timestamp
What proves value?	Protected test, baseline and uncertainty
Who may be harmed?	Subgroup performance and error consequences
When must use stop?	Warning and suspension triggers with owners

Recommendation

Treat the model as a controlled decision-support component. Advance only through a documented validation gate. Retain human authority, an override mechanism and monitoring that can trigger investigation or suspension. The strongest metric is not the smallest number; it is the estimate produced by the most credible operating experiment.

Canonical resources

- PU-B02-C01 Supervised Learning Foundations SRAI lesson
- M2_N01 supervised learning foundations executed notebook
- PU-B02-C01 lesson presentation with audience-facing narration
- Student assessment and instructor solutions